

DATA STRUCTURES

LAB PROGRAM

8. Write a C program to search a number using Linear Search method

```
#include <stdio.h>

int main()
{
    int arr[100], n, key, i, found = 0;

    printf("Enter number of elements: ");

    scanf("%d", &n);

    printf("Enter array elements:\n");

    for(i = 0; i < n; i++)

        scanf("%d", &arr[i]);

    printf("Enter number to search: ");

    scanf("%d", &key);

    for(i = 0; i < n; i++)
    {
        if(arr[i] == key)
        {
            found = 1;

            break;
        }
    }

    if(found)

        printf("Element found at position %d", i + 1);

    else

        printf("Element not found");
```

```
    return 0;  
}
```

Output

```
Enter number of elements: 5  
Enter array elements:  
10 20 30 40 50  
Enter number to search: 40  
Element found at position 4
```

```
=== Code Execution Successful ===
```